

Problem Set 5 Answers

1a. If you meet a person who used their preferred Generative AI, and the AI was correct, the probability that they are using ChatGPT is $\frac{(.50)(8/10)}{(.50)(8/10)+(.33)(9/10)+(.17)(13/15)} = 0.4737$.

1b. If you meet a person who used their preferred Generative AI, and the AI was correct, the probability that they are using Gemini is $\frac{(.33)(9/10)}{(.50)(8/10)+(.33)(9/10)+(.17)(13/15)} = 0.3518$.

1c. If you meet a person who used their preferred Generative AI, and the AI was correct, the probability that they are using Claude is $\frac{(.17)(13/15)}{(.50)(8/10)+(.33)(9/10)+(.17)(13/15)} = 0.1745$.

1d. If you pick a person at random, the probability that the AI will be incorrect, when they use their preferred Generative AI, is $(.50)(2/10) + (.33)(1/10) + (.17)(2/15) = 0.1557$.

2. There are exactly 21 equally-likely outcomes, and 6 of these have sum exactly 7, so the conditional probability that the sum of the dice is exactly 7 is $6/21 = 2/7$.

3. There are $(52)(51)$ equally-likely outcomes, $(52)(3)$ of which have the same values, so the probability that the values on the two cards agree is $((52)(3))/((52)(51)) = 3/51 = 1/17$.

An alternative view is that, no matter which card is the first one selected, 3 out of the remaining 51 cards (all of which are equally-likely to appear as the second card) have the same value as the first card. So the probability is $3/51 = 1/17$.

4. For $0 \leq k \leq 6$, to get the probability that Bob has k heads, we use a to denote the value on Alice's die. Then the probability that Bob has k heads is $\sum_{a=1}^6 (1/6) \binom{a}{k} (1/2)^a$ for $k = 0$, or $\sum_{a=k}^6 (1/6) \binom{a}{k} (1/2)^a$ for $1 \leq k \leq 6$, and therefore

4a. the probability Bob has 0 heads is $21/128$;

4b. the probability Bob has 1 head is $5/16$;

4c. the probability Bob has 2 heads is $33/128$;

4d. the probability Bob has 3 heads is $1/6$;

4e. the probability Bob has 4 heads is $29/384$;

4f. the probability Bob has 5 heads is $1/48$;

4g. the probability Bob has 6 heads is $1/384$.

5. As given in **4a**, the probability that Bob has 0 heads is:

$$(1/6)(1/2)^1 + (1/6)(1/2)^2 + (1/6)(1/2)^3 + (1/6)(1/2)^4 + (1/6)(1/2)^5 + (1/6)(1/2)^6 = 21/128.$$

Thus, given that Bob has 0 heads (i.e., all of his flips are tails), for $1 \leq k \leq 6$, the conditional probability that Alice rolled a value of k is $(1/6)(1/2)^k / (21/128) = (64/63)(1/2)^k$. Therefore:

5a. the conditional probability that Alice rolled a value of 1 is $32/63$.

5b. the conditional probability that Alice rolled a value of 2 is $16/63$.

5c. the conditional probability that Alice rolled a value of 3 is $8/63$.

5d. the conditional probability that Alice rolled a value of 4 is $4/63$.

5e. the conditional probability that Alice rolled a value of 5 is $2/63$.

5f. the conditional probability that Alice rolled a value of 6 is $1/63$.